MTH220

Timed Online Assignment – July Semester 2021

Statistical Methods and Inference

Thursday, 25 November 2021 **1:00 pm – 3:30 pm**

Time allowed: 2.5 hours

INSTRUCTIONS TO STUDENTS:

1. This Timed Online Assignment (TOA) contains **FOUR (4)** questions and comprises **SEVEN (7)** pages (including cover page).

2. You must answer **ALL** questions.

3. If you have any queries about a question, or believe there is an error in the question, briefly explain your understanding and assumptions about that question before attempting it.

4. You MUST **submit your answers via Canvas (similar to TMA submission) at the end time of this TOA** (as stated on this cover page). The 15 minutes grace period as shown on Canvas is strictly meant for technical issues encountered during submission. Thereafter, you will not be able to submit your answers and you will be considered as having withdrawn from the course. **No appeal will be allowed.**

5. You are to include the following particulars in your submission: TOA Course Code, your Full Name, and SUSS PI Number. Name your submission file as TOACourseCode_FullName_StudentPI. Use underscore and NOT Space. *Example: TOAXYZ_RaphaelLee_T1923161 (omit D/O, S/O).*

6. Your submission should consist of only one file and must not exceed 500MB in size. The file must be a PDF or Microsoft Word file saved in .doc or .docx format. All answers are to be typed. Flowcharts and graphs may be scanned or photographed and embedded in the PDF or Word file provided it does not exceed the file size limit of 500MB.

7. For answers which cannot be typed and require to be hand-written, ensure that question number is clearly stated on each page. All uploaded hand-written answers must be clear, readable and complete. Marks will not be awarded for un-readable or incomplete images.

8. To prevent plagiarism and collusion, your submission will be reviewed by Turnitin. The Turnitin report will only be made available to the marker and you will not be able to view it.

9. The University takes plagiarism and collusion seriously, and your Turnitin report will be examined thoroughly as part of the marking process.

10. Marks will only be awarded if FULL working is shown.

Answer all questions. (Total 100 marks)

Question 1

The number of hours that 10 persons studied for an English test and their scores on the test are given in Table Q1:

Hours studied (x)	**Test score out of 100 (y)**
4	31
9	58
10	65
14	73
5	37
7	44
12	60
22	91
1	21
17	84

Table Q1

(a) Draw a scatterplot for this dataset. Describe what you see.

(4 marks)

(b) Determine the equation of the least squares line that approximates the regression of the test scores on the number of hours studied.

Hint:

$$\sum x^2 = 1385, \sum xy = 6982, \sum y^2 = 36562$$

(8 marks)

(c) Predict the average test score of a person who studied 24 hours for the test. Comment on your findings.

(4 marks)

(d) Calculate the Spearman's correlation coefficient and comment on your findings.

(6 marks)

(e) The Spearman's correlation coefficient requires the Ranking of study hours and scores. Will there be any difference whether we choose to rank in Ascending (Smallest = Rank 1) or Descending order (Largest = Rank 1)?

(3 marks)

Question 2

Singapore has recently announced ambitious targets for the adoption of Electric Vehicles (EV), where EVs were expected to completely take over all petrol vehicles by 2040. EVs are a great choice for Singapore because our electricity comes from Natural Gas, which releases less greenhouse gases (eg CO_2) per unit energy, compared to the Petrol-based internal combustion engines in our cars today.

In an EV-test bed organized in Singapore, researchers were interested to understand how Air-Conditioning affects the total distance per full charge. 103 EVs and 103 independent volunteers were recruited to test the vehicles where 56 will drive with aircon on, and 47 will drive with aircon off.

Table Q2 shows summary statistics obtained from the test-bed.

Total distance travelled on full charge (km)	
Aircon is ON (X) $\boldsymbol{n = 56}$	**Aircon is OFF (Y)** $\boldsymbol{n = 47}$
$\sum(X - 348) = 0$ $\sum(X - 348)^2 = 7523.946$	$\sum(Y - 386) = 0$ $\sum(Y - 386)^2 = 6321.277$

<u>Table Q2</u>

It is hypothesized that turning on the air-con will cause the battery to support 33.4 km less total distance per full charge.

(a) Suggest a suitable parametric hypothesis test, stating the null and alternative hypothesis clearly.

(3 marks)

(b) At 5 % level of significance, apply a hypothesis test to determine if turning on the aircon will result in 33.4km less total distance per full charge. Comment on the results of the hypothesis test.

(10 marks)

(c) State the assumptions you made in Question 2(b) for you to perform the hypothesis testing. Verify if these assumptions were met.

(3 marks)

(d) Construct a 95 % confidence interval for the difference in the mean distance.

(5 marks)

(e) In statistical hypothesis testing, there are also non-parametric approaches which we can use instead.

(i) Explain when we prefer to use non-parametric test instead of a parametric test.

(4 marks)

(ii) Explain why we are unable to use the Wilcoxon Signed Rank Test for this dataset and suggest how can the experiment be altered to produce data suitable for a Wilcoxon Signed Rank Test.

(5 marks)

Question 3

Suppose that we have a normal population with known population variance $\sigma^2 = 1$. We took a sample of n = 20 and obtained a sample mean $\bar{x} = 5$.

(a) Apply a one-tailed Z-test at $\alpha = 0.05$ to test the null hypothesis $\mu_0 = 4.5$ against the alternative hypotheses $\mu_0 > 4.5$ and show that there is sufficient evidence to reject the null hypothesis.

(6 marks)

(b) Assume H_0 is **FALSE,** that the real population mean $\mu_0 = 5$ instead of $\mu_0 = 4.5$. Compute β, the probability of getting $\bar{x} < 4.8678$. (Hint: This is the probability of a Type-2 error.)

(4 marks)

(c) Explain the difference between Type-1 and Type-2 errors.

(4 marks)

(d) Figure Q3(d) shows the Type-1 error in black and Type-2 error in red for different cutoff values. For example, the left plot in Figure Q3(d) shows $x = 4.8678$ and the right plot in Figure Q3(d) shows $x = 5$. Describe the relationship between the probabilities α and β.

Can we minimize the error, both α and β, of a hypothesis test?

(6 marks)

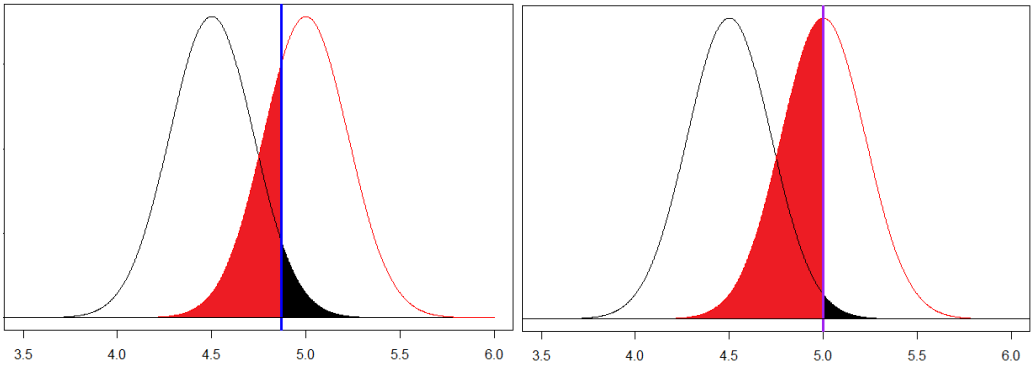

Figure Q3(d)

Question 4

(a) A fair octahedral die, with faces marked 1, 2, 3, 4, 5, 6, 7, 8 is thrown 90 times. Let X be the number of times that a "3" is obtained.

(i) What is the sampling distribution of X? Calculate $E(X)$ and $Var(X)$.

(3 marks)

(ii) Use a suitable approximation to find the probability that a 3 is obtained fewer than 10 times. Comment on your use of the approximation.

(5 marks)

(b) A study of marital status and education level in Singapore was conducted by sending surveyors to all parts of Singapore. Great care is done to ensure that people were randomly selected, and citizens of all socioeconomic status were represented.

The researchers managed to get 2783 respondents aged 30 to 54 at the point of the survey. All of them are people born in Singapore and have stayed in Singapore for most of their life.

Based on the results tabulated in Table Q4(b), we would like to find out whether there is any statistical evidence that the data show a difference in marital status by education level.

	Single	**Married**	**Widowed Divorced Separated**
Below Secondary	99	623	177
Secondary and Post-Secondary	224	555	55
Diploma, Professional Qualification, University	356	662	32

Table Q4(b)

(i) Formulate the null and alternative hypotheses.

(2 marks)

(ii) Perform a chi-squared test at 5 % level of significance to determine whether there is any association between gender and verbal reasoning ability. Comment on the results of the hypothesis test.

(11 marks)

(iii) Critically evaluate the study design and the analysis performed. How can you improve on the study?

(4 marks)

----- END OF PAPER -----